

Business Requirements Document (BRD)

Multi-Channel Ad Spend Efficiency & Funnel Leakage Analysis

Business Objective

The business needs a **performance tracking report** for advertising campaigns running on **Facebook and Instagram**.

The report will provide visibility into campaign reach, engagement, conversions, and budget utilization.

This will enable the marketing team to:

- Identify the most effective platform (Facebook vs Instagram).
 - Track campaign ROI and optimize budget allocation.
 - Understand audience engagement patterns.
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Scope of the Report

In Scope:

- Campaigns running on **Facebook and Instagram** only.

Out of Scope:

- Other platforms (Messenger, Audience Network).
 - Organic engagement (only **paid ads** will be included).
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KPIs & Definitions

KPI	Definition	Formula	Example Use
Impressions	Number of times ads were displayed.	Count of event_type = Impression	Measure reach
Clicks	Number of times users clicked ads.	Count of event_type = Click	Measure engagement intent
Shares	Number of times ads were shared.	Count of event_type = Share	Viral engagement
Comments	Number of user comments on ads.	Count of event_type = Comment	User sentiment & feedback
Purchases	Number of purchases made after seeing ads.	Count of event_type = Purchase	Conversions

KPI	Definition	Formula	Example Use
Engagements	Total interactions (Clicks + Shares + Comments).	Clicks + Shares + Comments	Engagement volume
CTR (Click Through Rate)	% of impressions that resulted in clicks.	$(\text{Clicks} \div \text{Impressions}) \times 100$	Ad effectiveness
Engagement Rate	% of impressions that resulted in engagements.	$(\text{Engagements} \div \text{Impressions}) \times 100$	Overall ad appeal
Conversion Rate	% of clicks that resulted in purchases.	$(\text{Purchases} \div \text{Clicks}) \times 100$	Funnel efficiency
Purchase Rate	% of impressions that resulted in purchases.	$(\text{Purchases} \div \text{Impressions}) \times 100$	Conversion from reach
Total Budget	Total spend allocated to campaigns.	Sum of campaigns.total_budget	Cost analysis
Avg. Budget per Campaign	Average budget allocation per campaign.	$\text{Total Budget} \div \text{Campaign Count}$	Budget distribution

Charts Requirements

1. Target Gender — Donut Chart

A **donut chart** will visualize performance split by **target gender** (from the ads table).

- The metric displayed (e.g., Impressions, Clicks, Purchases) will change dynamically via the parameter.
- **Purpose:** Identify which gender segment contributes most to the selected metric.

2. Target Age Group — Bar Chart

A **bar chart** will show engagement across **age groups** defined in the ads table.

- Each bar will represent one age group.
- The metric displayed will switch dynamically.
- **Purpose:** Highlight which age group is most responsive to campaigns.

3. Country — Map

A **map visualization** will display performance by **country** (from the users table).

- Bubble size or color intensity will represent the selected metric.
- **Purpose:** Provide a geographic view of campaign reach and engagement.

4. Calendar Month — Calendar Heat Map

A **calendar heat map** will plot performance at the **monthly level**, based on the timestamp field in `ad_events`.

- Darker shades will indicate higher activity.
- **Purpose:** Detect seasonal trends, peak ad months, and low-activity periods.

5. Weekly Trend — Stacked Column by Ad Type

A **stacked column chart** will display weekly performance trends.

- X-axis → Week number (from the Date Table linked to `ad_events`).
- Stacks → Different `ad_type` values (from the ads table).
- Y-axis → Selected metric.
- **Purpose:** Compare ad type contributions over weeks.

6. Hourly Trend — Area Chart

An **area chart** will show activity by **hour of day** (from `ad_events[time_of_day]`).

- X-axis → Hour of the day (0–23).
- Y-axis → Selected metric.
- **Purpose:** Understand user activity patterns throughout the day.

7. Ad Type — Matrix

A **matrix visualization** will show the selected metric across **ad types** and break down further by **platform (Facebook vs Instagram)**.

- Rows → Ad Types.
- Columns → Platforms or other campaign dimensions.
- Values → Selected metric.
- **Purpose:** Compare performance across ad formats and platforms side by side.